

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that: when a DNS request is sent to the DNS server the port is unreachable.

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: udp port 53 unreachable

The port noted in the error message is used for: the port 53 is mostly used for DNS service

The most likely issue is: from the dns server. The port 53 which has the domain ip address is unreachable or has some kind of firewall configuration issue.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred: The incident happened in the afternoon around 1:24pm.

Explain how the IT team became aware of the incident: customers couldn't have access to the website with that port number so they filed a complaint which the analyst too tried reaching the site and also received the same error message they received.

Explain the actions taken by the IT department to investigate the incident: they first tried to confirm the issue by accessing the same site which they got the same error message and then went ahead to troubleshoot the issue by loading the network analyzer tool, tcpdump, and attempted reloading the web page again.

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.): After the reloading of the webpage again with the help of the network protocol analyzer, the analyzer revealed that when UDP packets are sent to the DNS server, the ICMP packets send an error message saying "udp port 53 unreachable"

Note a likely cause of the incident: The clear cause for now is that the DNS service is unavailable on the udp port 53 or maybe some firewall rule blocking access to the udp

port 53.